

Homework #2 (Submit before 30-Mar)

❖ Derive the ray matrices for cases (1) to (5) on next page.

❖ Ch2_No.3

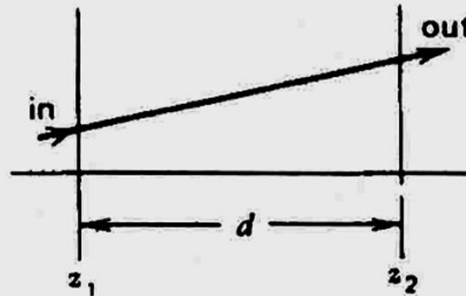
Based on the matrix formulation for the light propagation, prove that the confocal cavity is a stable cavity, that is, any paraxial light can go back and forth within it multiple times, and it will form close-loop after two rounds.

❖ Ch2_No.5

The resonant cavity of the laser is composed of a convex mirror with a curvature of 1 m and a concave mirror with a curvature of 2 m, respectively. The working substance is 0.5 m long with refractive index of 1.52. Please determine the range of cavity length L to form a stable cavity.

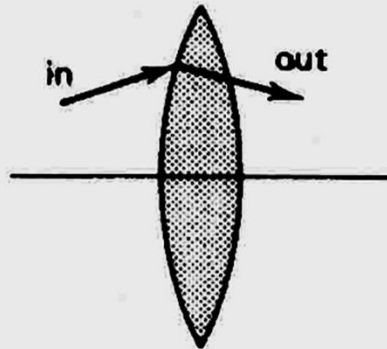
Ray matrices for some common cases

- (1) Homogeneous Medium:
Length d



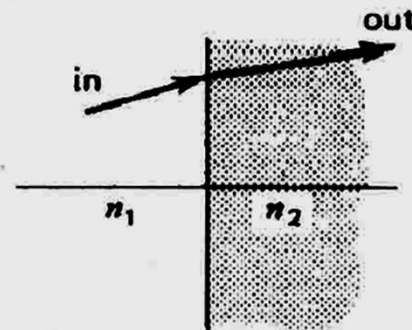
$$\begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$$

- (2) Thin Lens:
Focal length f
($f > 0$, converging;
 $f < 0$, diverging)



$$\begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$

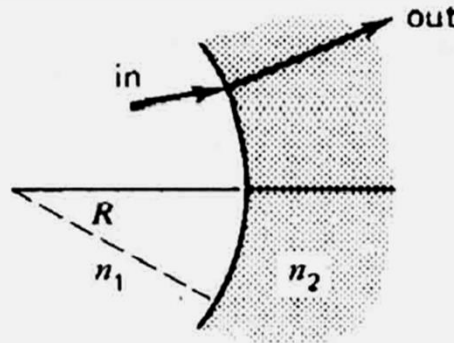
- (3) Dielectric Interface:
Refractive indices
 n_1, n_2



$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}$$

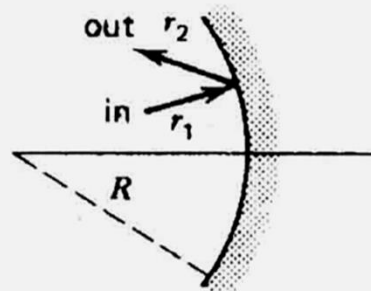
Ray matrices for some common cases

- (4) Spherical Dielectric Interface:
Radius R



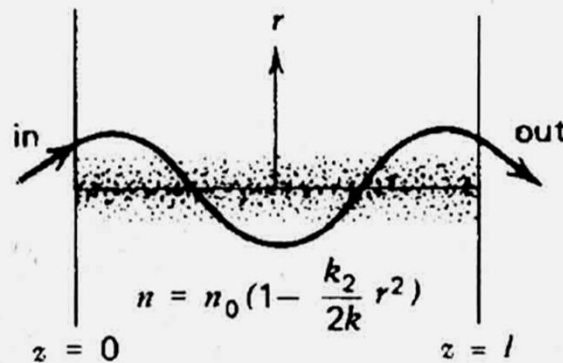
$$\begin{bmatrix} 1 & 0 \\ \frac{n_2 - n_1}{n_2} \frac{1}{R} & \frac{n_1}{n_2} \end{bmatrix}$$

- (5) Spherical Mirror:
Radius of curvature R



$$\begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix}$$

- (6) A medium with a quadratic index profile



$$\begin{bmatrix} \cos\left(\sqrt{\frac{k_2}{k}} l\right) & \sqrt{\frac{k}{k_2}} \sin\left(\sqrt{\frac{k_2}{k}} l\right) \\ -\sqrt{\frac{k_2}{k}} \sin\left(\sqrt{\frac{k_2}{k}} l\right) & \cos\left(\sqrt{\frac{k_2}{k}} l\right) \end{bmatrix}$$

平方折射率介质